

Module 1

Explaining Network Topologies

Learning Objectives

- Explain network types and characteristics.
- Compare and contrast OSI model layers.
- Configure SOHO networks.
- Explain CompTIA's troubleshooting methodology.

Lesson 1.1

Networking Overview

Think About It

How would you define your social network?



How are Personal and IT Networks Similar?

Social Network

A collection of friends, family, coworkers, and acquaintances

Provides emotional support, advice, and opportunities in personal matters

Helps in communication, information sharing, and social connections

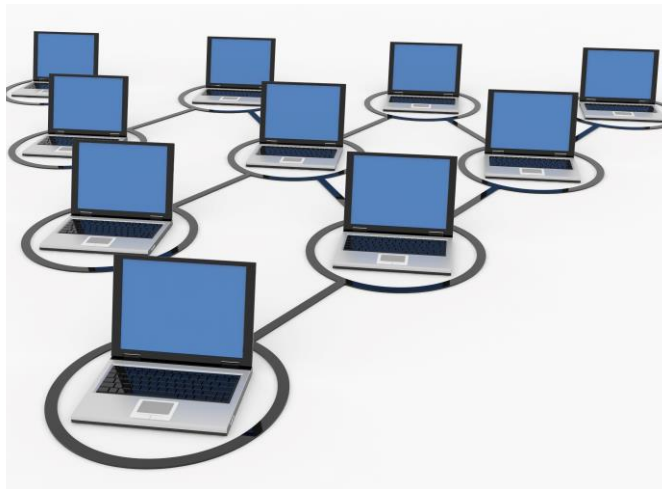
IT Network

A collection of computers and other devices connected by transmission medium

Provides an opportunity for shared resources

Helps with communication and information sharing in a digital environment

Networks



Peer-to-Peer



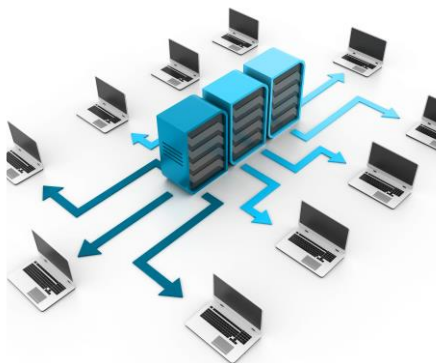
Client-Server

Local Area Networks

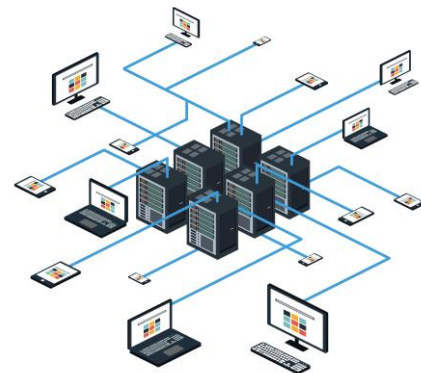
A local area network (LAN) is confined to a single geographical location.



SOHO



Enterprise LAN



Datacenter

Activity: Quick Search

Point-to-Point
Topology

Star Topology

Mesh Topology



Network Topologies: Point-to-Point

1

In a point-to-point (or duplex) network, only two nodes are connected to the network media.



2

When the network media is half-duplex, a node cannot transmit and receive at the same time.

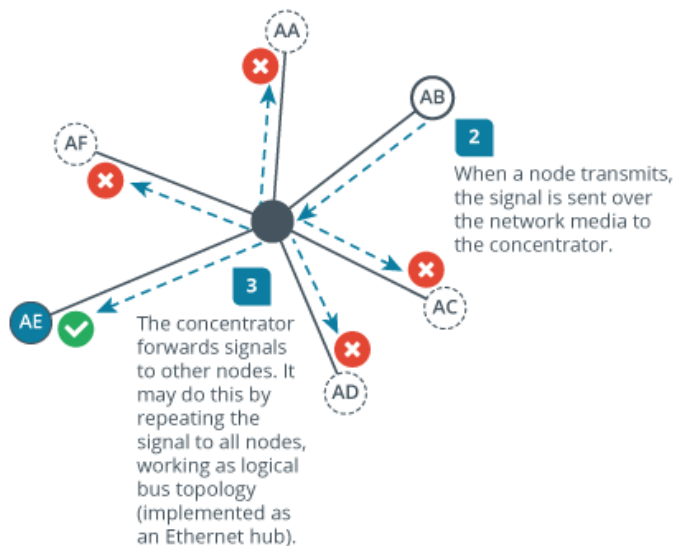


3

When the network media is full-duplex, nodes can transmit and receive simultaneously.

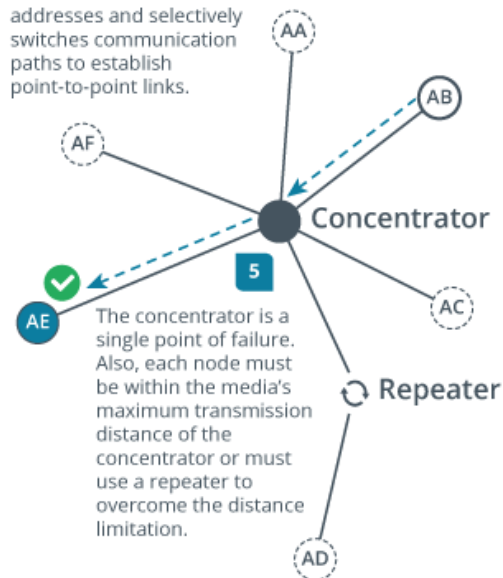
Network Topologies: Star

- 1 In a star topology, each node is connected to a concentrator over dedicated network media.



4

More commonly, the concentrator tracks node addresses and selectively switches communication paths to establish point-to-point links.

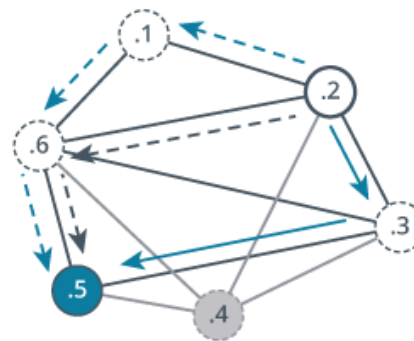


Network Topologies: Mesh

- 1** In a fully connected mesh network, each node has a point-to-point link with every other node. This requires exponentially more links as nodes are added: $n*(n-1)/2$



- 2** Provisioning so many interfaces and links is difficult, so partial mesh networks are often preferred. In a partial mesh, nodes can forward packets to a destination by learning the network topology.



- 3** Packets can take multiple routes through the network, providing resilience if some nodes or links fail.

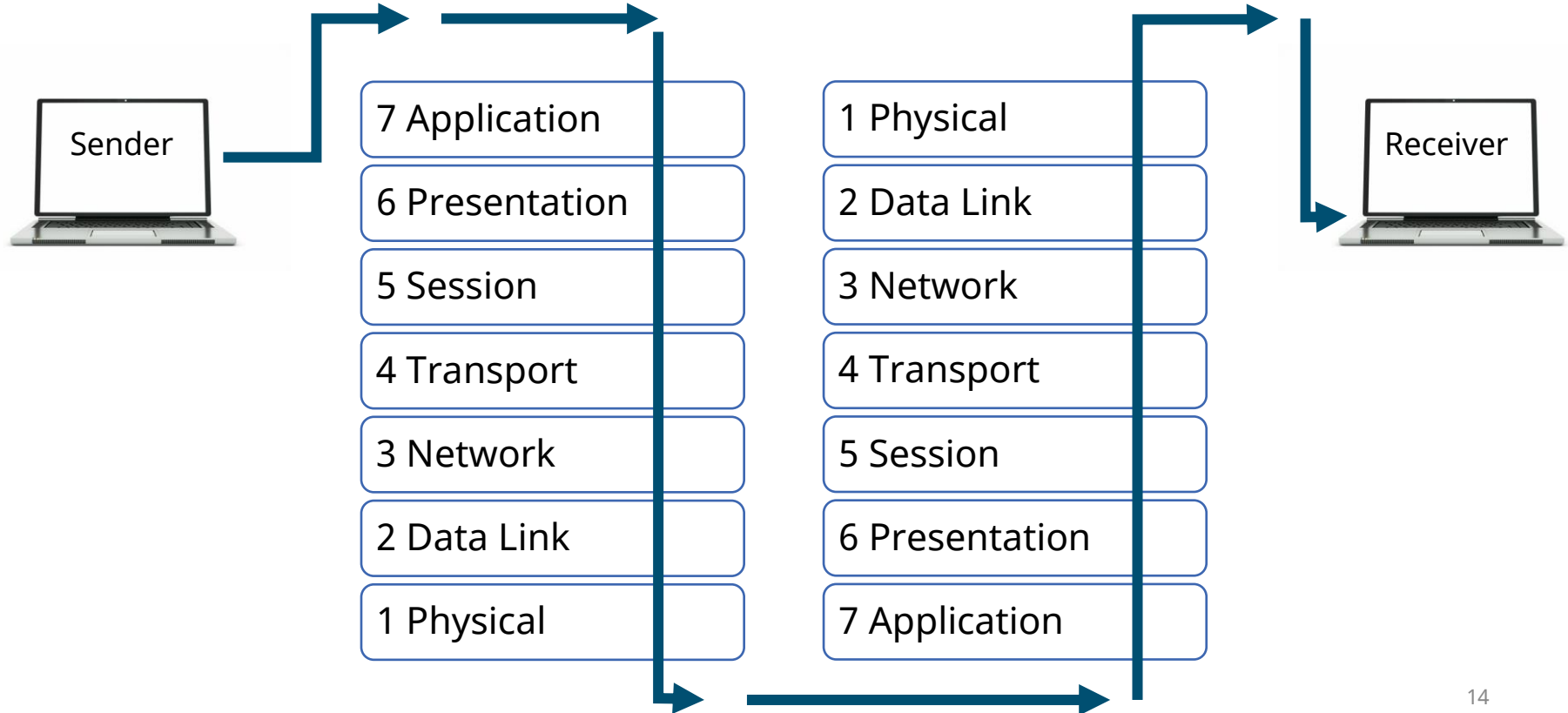
Lesson Summary

- **Network:** Connected devices sharing data (computers, protocols, transmission medium)
- **Client-Server Network:** Central server provides resources and services to client devices
- **Local Area Network (LAN):** Connects devices within a limited area (home, office)
- **Network Topology:** Layout of connections between network devices (star, mesh, point-to-point)

Lesson 1.2

OSI Model Concepts

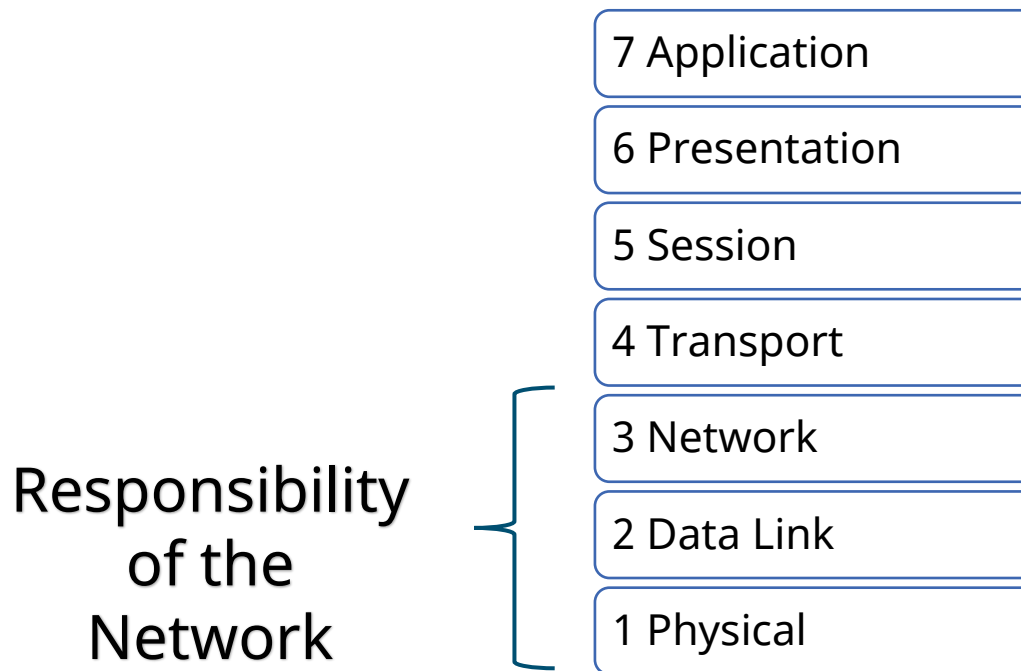
The OSI Model Packet Flow



The OSI Model Mnemonic

Please	Do	Not	Throw	Sausage	Pizza	Away
P	D	N	T	S	P	A
Physical	Data Link	Network	Transport	Session	Presentation	Application

Responsibility of the Network



Layer 1 – Physical

7 Application

6 Presentation

5 Session

4 Transport

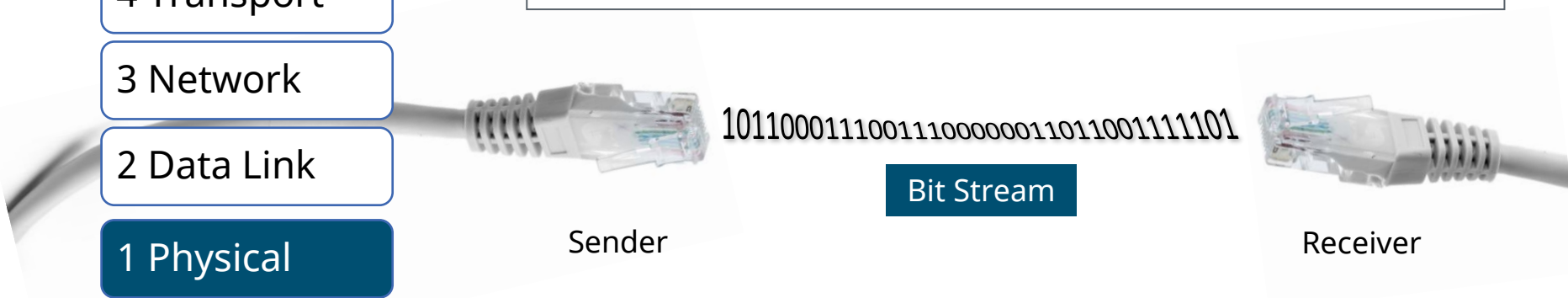
3 Network

2 Data Link

1 Physical

Functions and Examples

- Transmit bits from sending source over network communications to physical layer of receiving device
- Data transmitted using cables or wireless media
- Examples: Coax and fiber cables, hubs and repeaters, modem, transceiver



Layer 2 – Data Link

7 Application

6 Presentation

5 Session

4 Transport

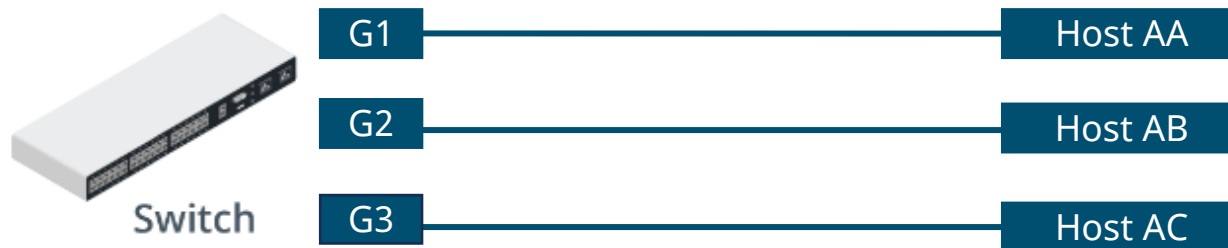
3 Network

2 Data Link

1 Physical

Functions and Examples

- Transfers data between nodes on the same logical segment using hardware addresses
- Encapsulation: organizes stream of bits arriving from the Physical layer into frames
- Examples: NIC, bridge, switch, wireless access point



Layer 3 – Network

7 Application

6 Presentation

5 Session

4 Transport

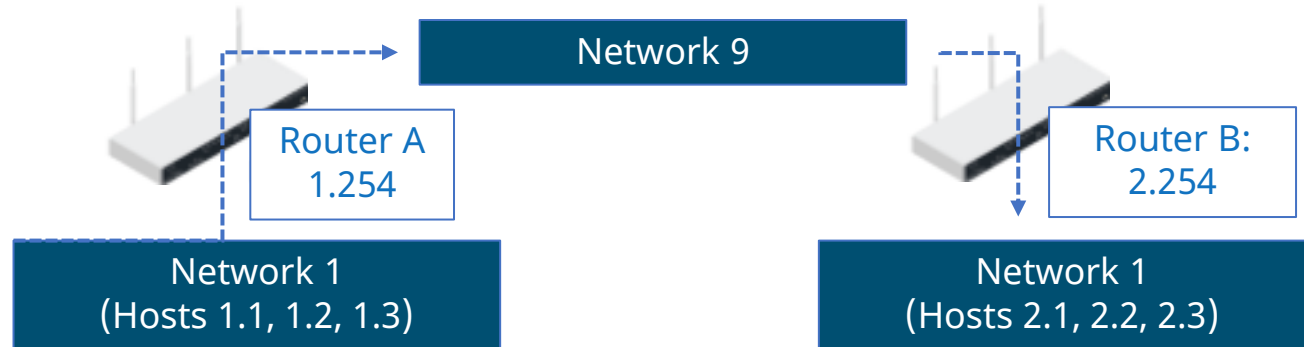
3 Network

2 Data Link

1 Physical

Functions and Examples

- Moving data around a network of networks using network and host IDs
- Packets given a destination IP address and forwarded to the destination network
- Examples: Router, IP, ACL, Basic Firewall



Layer 4 - Transport

7 Application

6 Presentation

5 Session

4 Transport

3 Network

2 Data Link

1 Physical

Functions and Examples

- Tracks communication between applications on source and destination hosts
- Segments data and manages each data piece
- Reassembles segments into application data
- Examples: Multilayer switches, advanced firewalls, intrusion detection systems (IDSs)

CompTIA.



Responsibility of Host

Responsibility of the Host

7 Application

6 Presentation

5 Session

4 Transport

3 Network

2 Data Link

1 Physical

Layer 5 - Session

7 Application

6 Presentation

5 Session

4 Transport

3 Network

2 Data Link

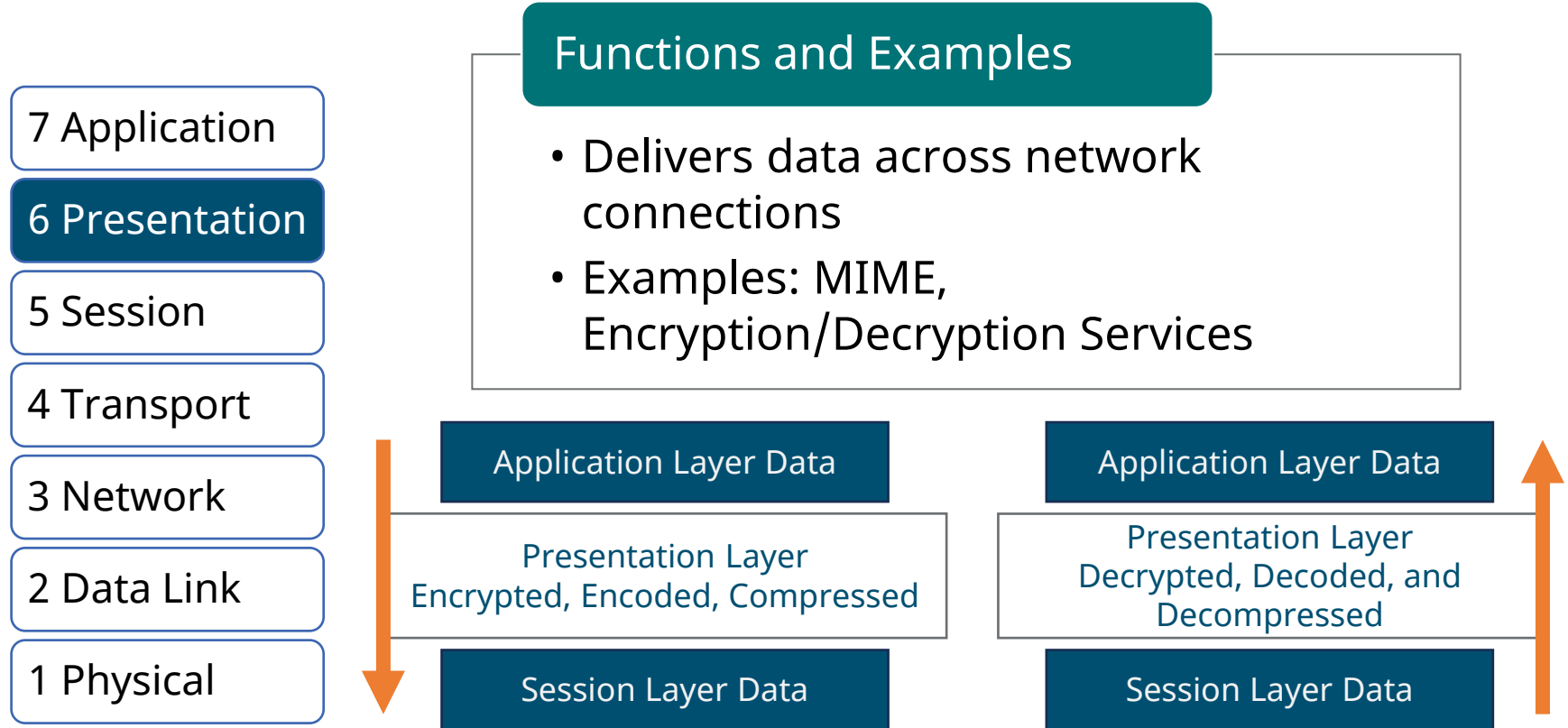
1 Physical

Functions and Examples

- Establishes and manages the connections between applications
- Examples: ASP, ADSP, NetBIOS, PAP



Layer 6 - Presentation



Layer 7 - Application

7 Application

6 Presentation

5 Session

4 Transport

3 Network

2 Data Link

1 Physical

Functions and Examples

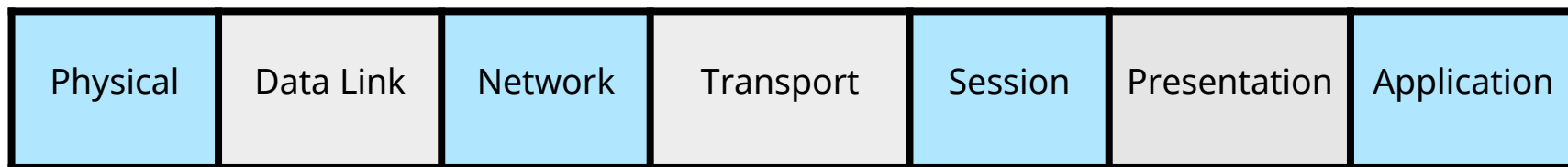
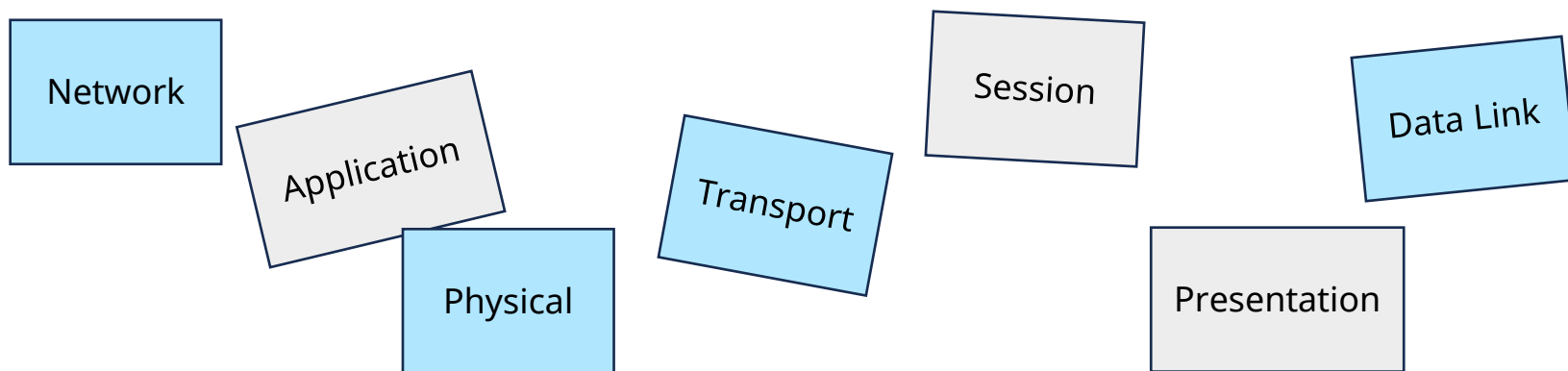
- Interacts directly with end-user's software applications for network service
- Examples: SMTP, DNS, FTP



Human
Readable

Machine
Readable

Activity: Unscramble the OSI Model



Lesson 1.3

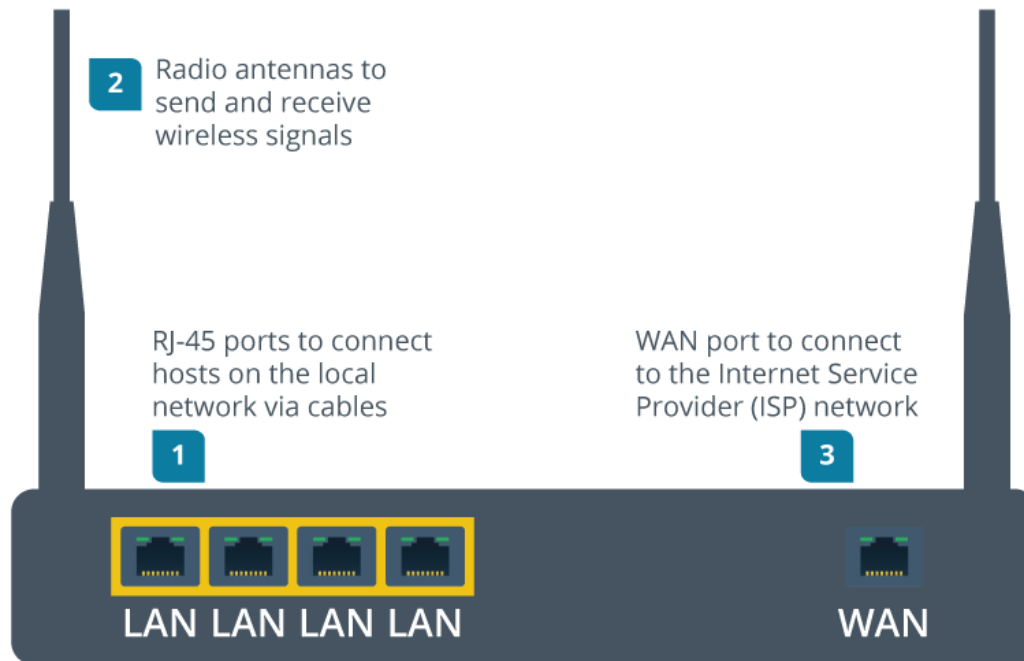
SOHO Networks

Think About It

What is a SOHO router?
And what is it used for?

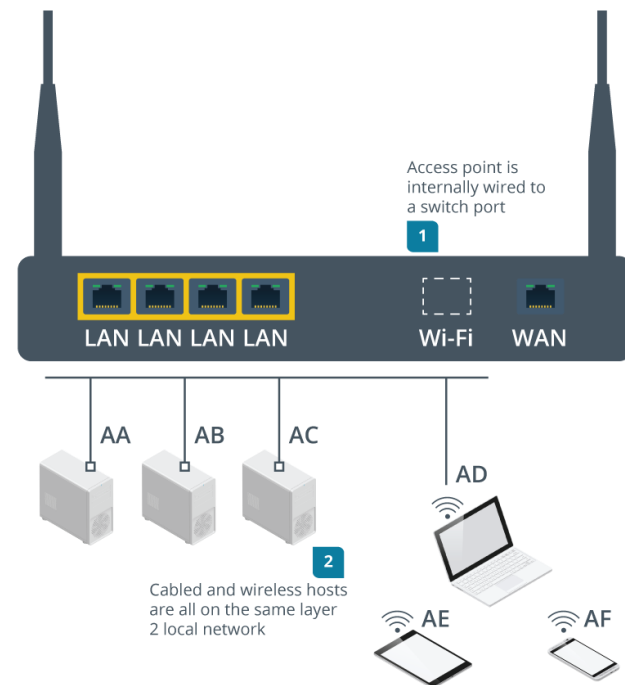


SOHO Router – Physical Layer



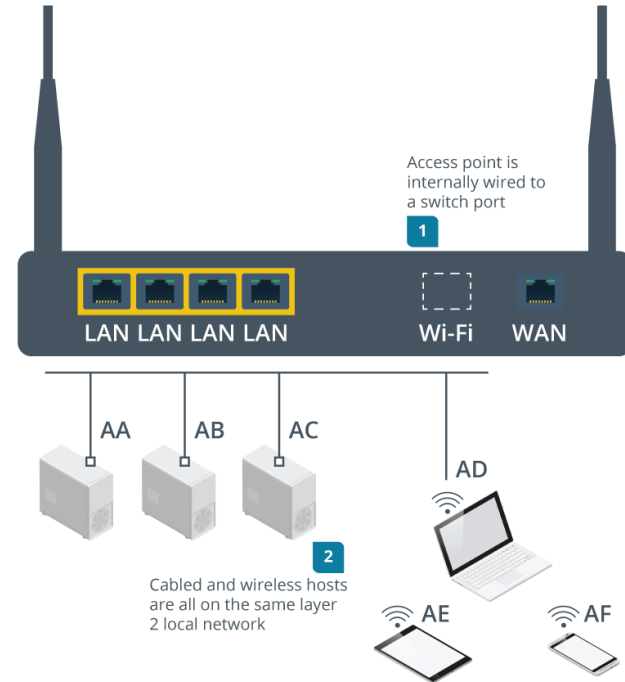
SOHO Router – Data Link Layer

- Ethernet switch
- Wireless access point
- Each host interface identified by a MAC address

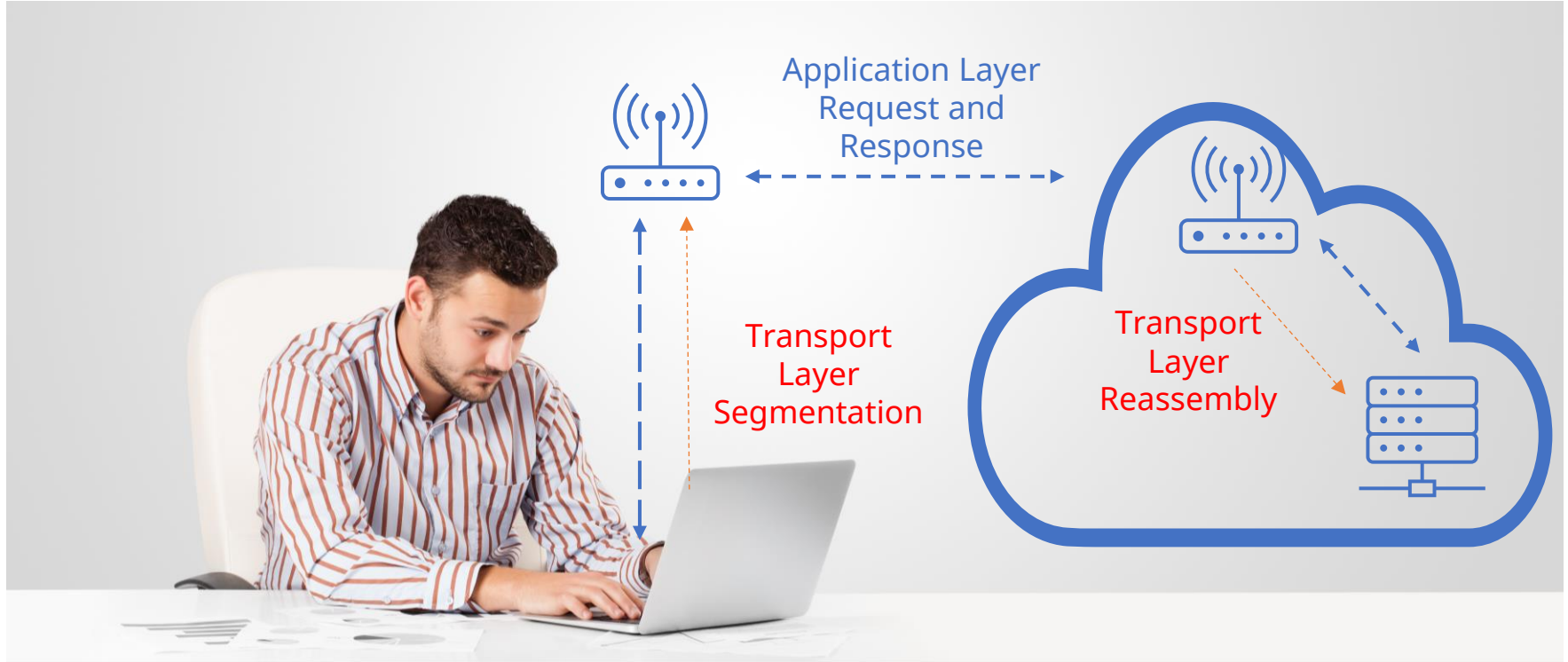


SOHO Router – Network Layer

- Forwards between local private network and public internet
- Runs a DHCP server to provide each host with an IP address
- Router's WAN interface given a public IP address



SOHO Router – Transport and Application



SOHO Router – The Internet

- WAN: SOHO connection to Internet
- Public switched telephone network (PSTN)
- High-bandwidth trunks connecting IXPs
- ISP links between networks in IXP datacenter



Binary Numbers – Base 10

Ten-millions	One-millions	Hundred-thousands	Ten-thousands	One-thousands	Hundreds	Tens	Ones
Digit x 10,000,000	Digit x 1,000,000	Digit x 100,000	Digit x 10,000	Digit x 1,000	Digit x 100	Digit x 10	Digit x 1

Binary Numbers – Base 10

128	64	32	16	8	4	2	1
Digit x 128	Digit x 64	Digit x 32	Digit x 16	Digit x 8	Digit x 4	Digit x 2	Digit x 1

Binary Conversion

Let's look at how to convert 205 into binary.

128	64	32	16	8	4	2	1
Digit x 128	Digit x 64	Digit x 32	Digit x 16	Digit x 8	Digit x 4	Digit x 2	Digit x 1
1	1	0	0	1	1	0	1

Activity: Binary Conversion

Convert 132 into binary.

128	64	32	16	8	4	2	1
Digit x 128	Digit x 64	Digit x 32	Digit x 16	Digit x 8	Digit x 4	Digit x 2	Digit x 1

Activity: Binary Conversion

Convert 132 into binary.

128	64	32	16	8	4	2	1
Digit x 128	Digit x 64	Digit x 32	Digit x 16	Digit x 8	Digit x 4	Digit x 2	Digit x 1
1	0	0	0	0	1	0	0

Lesson 1.4

Troubleshooting Methodology

Discussion: Think About It

What is troubleshooting?

What steps should be involved in the troubleshooting process?



Troubleshooting Methodology



Identify the Problem

Gather information

- System documentation
- Installation and maintenance logs
- Vendor support sites

Question users

- Open questions to uncover information
- Closed question to invite a yes or no or fixed response



Identify Problem Symptoms

Identify symptoms

- Physical inspection
- Logs and diagnostic software
- Try to duplicate the problem

Determine if anything has changed

- Did it ever work?
- What has changed since it was last working?

Approach problems individually

- Verify symptoms are related before treating them as one single issue



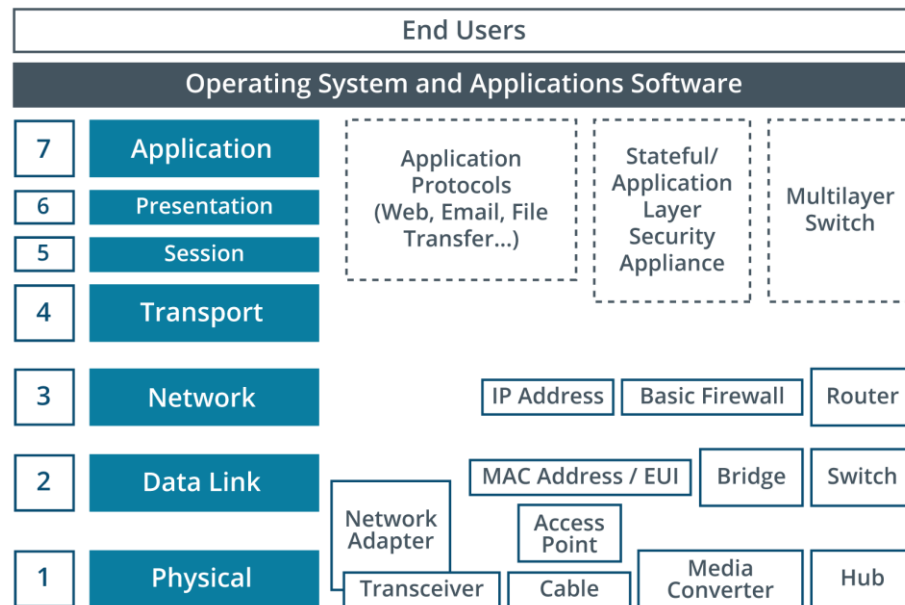
Establish a Theory of Probable Cause

Troubleshooting styles

- Question the obvious
- Methodically prove the functionality of each component

OSI Model Approach

- Top-down approach
- Bottom-up approach



Test the Theory to Determine the Cause

Question the obvious

- Use one or more method-based approaches
- Don't jump to conclusions

Escalation

- Problem is beyond your knowledge or ability
- Problem falls under a system warranty
- Scope of the problem is very large
- Customer is difficult or abusive



Implement, Verify, and Document

Implement changes

- Implement fixes
- Escalate if higher authorization is needed

Verify the solution

- Make sure the system seems to work for you
- Make sure the system seems to work for customer

Document

- Document findings, actions, and outcomes
- Logs are useful for future troubleshooting



Summary

- Remember OSI model's 7 layers (physical, data link, network, transport, session, presentation, and application)
- Plan network structure and troubleshoot issues using OSI model layers
- Implement methodical approach to troubleshooting (identification, planning, execution, verification, and documentation)
- Use strategies like top-to-bottom, bottom-to-top, and divide-and-conquer